JAIN FACULTY OF ENGINEERING AND TECHNOLOGY

Department of Computer Science & Engineering

Global Campus, Jakkasandra Post, KanakapuraTaluk, Ramanagara District, Pin Code: 562 112

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

PCL Attendance

Team No. : COEPCL-0177

Project Title : AI-Powered Smart Reverse Vending Machine for Automated Waste Classification and Reward System.

Sl. No.	Date of the Meeting	Points Discussed	Members Absent (Roll No)	Signature of the Mentor
1.	05/01/26	Project idea discussion and domain finalization for AI based waste classification and smart recycling system.	00/05	[signature]
2.	12/01/26	Collected research papers and explored ML approaches for image classification. finalized transfer learning as per preferred method.	00/05	[signature]
3.	19/01/26	Discussed dataset sources from kaggle and Roboflow. finalized waste categories & explored ResNet based classification architecture.	00/05	[signature]
4.	27/01/26	Started dataset collection and preprocessing. images were organized into several categories.	00/05	[signature]
5.	02/02/26	Conducted initial model training experiments using transfer learning techniques. performed validation testing and analyzed early prediction outputs.	00/05	[signature]

6.	10/02/26	improved dataset quality and class balance. Tested image augmentation and optimized training parameters for better accuracy.	00/05	
7.	17/02/26	Finalized six class classification structure for practical RVM implementation. conducted confusion matrix analysis and validation testing.	00/05	
8.	23/02/26	Started frontend development using HTML, CSS, JS. implemented image upload functionality and prediction display interface.	00/05	
9.	04/03/26	integrated Flask backend with trained ml learning model. API communication between frontend and backend tested successfully.	00/05	
10.	10/03/26	implemented Docker based deployment for isolated model hosting. integrated ONNX runtime for optimized interface performance	00/05	
11.	18/03/26	Enhanced UI/UX with dark mode, statistics dashboard, responsive layout and scan history functionality.	00/05	
12.	24/03/26	completed final model using 8.7K images approx and achieved 96.7 validation accuracy conducted testing under multiple environments	00/05	